ANDROID PROGRAMMING PRACTICAL 08

Module 2 · Notifications using NotificationManager

Aim

Implement notifications using NotificationManager.

Project name: NotificationDemo

Requirements

Android Studio, Java, Android SDK, and an Android Virtual Device or compatible Android phone. Use the package name generated by your project and adjust imports/dependencies to the installed Android Studio template.

Procedure

1. Create a project and add a button to trigger a notification.

2. For Android 8.0/API 26 and later, create a notification channel with NotificationManager.

3. Add the POST_NOTIFICATIONS runtime permission flow for Android 13/API 33 and later where required.

4. Build a notification using NotificationCompat.Builder with a small icon, title, text, and priority/channel.

5. Obtain NotificationManagerCompat and call notify() with a stable notification ID.

6. Run on an emulator, grant permission if requested, and tap the button.

7. Verify the notification appears in the notification shade; test dismissal.

Implementation checklist

• Keep resource names lowercase and use unique view IDs.

• Make sure every referenced layout, menu, drawable, string, or Java class exists in the project.

• Build the project and resolve compile errors before running it.

• Use Logcat to diagnose runtime errors and permissions or manifest issues.

Expected result

A notification is shown in the notification shade when permissions and channel requirements are met.

Testing / observations

Record the device/API level, the actions tested, the observed output, and any errors corrected. Attach a screenshot of the successful run to your practical record if required by your institution.

Conclusion

The stated Android concept was implemented and verified through the steps and tests above.